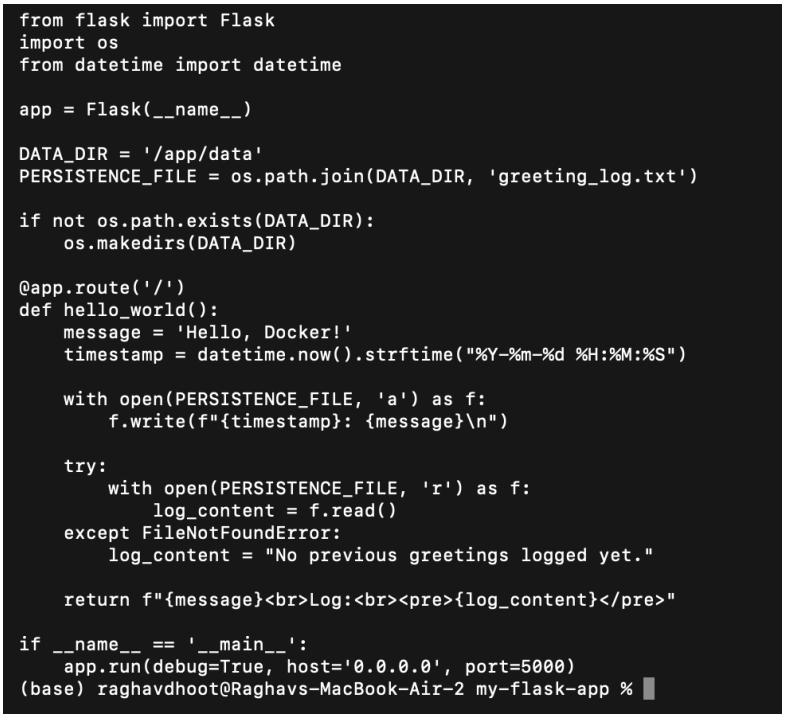
SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
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॥वसुधैव कुटुम्बकम्॥ Re-Accredited by NAAC with 'A' Grade

Department of Computer Science and Engineering
DevOps Lab - L3

Lab Assignment No: 10	
Name of Student	Raghav Dhoot
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Batch	2023 - 2027
Class	CSE - B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Docker Volumes & Data Persistence
Implementation & Theory	 app.py updated — Modified the Flask app to write a timestamped log entry to /app/data/greeting_log.txt on every request, and read back the accumulated log to display in the response — setting up a testable persistence check.

```
from flask import Flask
import os
from datetime import datetime

app = Flask(__name__)

DATA_DIR = '/app/data'
PERSISTENCE_FILE = os.path.join(DATA_DIR, 'greeting_log.txt')

if not os.path.exists(DATA_DIR):
    os.makedirs(DATA_DIR)

@app.route('/')
def hello_world():
    message = 'Hello, Docker!'
    timestamp = datetime.now().strftime("%Y-%m-%d %H:%M:%S")

    with open(PERSISTENCE_FILE, 'a') as f:
        f.write(f"{timestamp}: {message}\n")

    try:
        with open(PERSISTENCE_FILE, 'r') as f:
            log_content = f.read()
    except FileNotFoundError:
        log_content = "No previous greetings logged yet."

    return f"{message}<br>Log:<br><pre>{log_content}</pre>"

if __name__ == '__main__':
    app.run(debug=True, host='0.0.0.0', port=5000)
(base) raghavdhoot@Raghavs-MacBook-Air-2 my-flask-app %
```

```
Uttkarshs-Macbook-Air:my-flask-app uttkarshruparel$ docker build -t my-flask-app:v2 .
[+] Building 1.7s (10/10) FINISHED                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                0.0s
 => => transferring dockerfile: 194B                                                                                                0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim                                                                  1.5s
 => [internal] load .dockerignore                                                                                                   0.0s
 => => transferring context: 2B                                                                                                     0.0s
 => [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b            0.0s
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b            0.0s
 => [internal] load build context                                                                                                   0.0s
 => => transferring context: 1.07kB                                                                                                 0.0s
 => CACHED [2/5] WORKDIR /app                                                                                                       0.0s
 => CACHED [3/5] COPY requirements.txt .                                                                                            0.0s
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                                 0.0s
 => [5/5] COPY . .                                                                                                                  0.0s
 => exporting to image                                                                                                              0.1s
 => => exporting layers                                                                                                             0.0s
 => => exporting manifest sha256:cb26008b054d560c8cd6c0dcf321968cb0731b805cd333151f9dd5113ff3998e                                   0.0s
 => => exporting config sha256:a3d55d055b28c00d85a0ab030a6603ade87b53dfc947e4ae8cf9bfb5933f49ef                                     0.0s
 => => exporting attestation manifest sha256:f934bf38325d11a77752b2d4c5b3e2238abe667cdd21d1339ee1abc8f978b5e3                       0.0s
 => => exporting manifest list sha256:f0cd10e65847d6b55dc21b9c0d68662929c82ac8001701c5445061f502825023                              0.0s
 => => naming to docker.io/library/my-flask-app:v2                                                                                  0.0s
 => => unpacking to docker.io/library/my-flask-app:v2                                                                               0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/yojkxn36smsb3h7qwa76e8l00
```

Image rebuilt — Built a new image tagged my-flask-app:v2 incorporating the updated app code.

```
my-app-data
DRIVER    VOLUME NAME
local     my-app-data
(base) raghavdhoot@Raghavs-MacBook-Air-2 my-flask-app %
```

Named volume created — Created a Docker volume my-app-data; verified its existence with docker volume ls --filter name=my-app-data.

```
(base) raghavdhoot@Raghavs-MacBook-Air-2 my-flask-app % docker run -d -p 5001:5000 -v my-app-
data:/app/data my-flask-app:v2
89b5fee5562d9fefa95240656aa4264297b8cf7f3561c106eb437d10912e9dfe
```

Container run with volume mounted — Started a detached container from my-flask-app:v2, mounting my-app-data to /app/data inside the container.

> Hello, Docker!
> Log:
>
> 2026-10-12 11:42:50: Hello, Docker!

Verify writes accumulate — Refreshed http://localhost:5001 multiple times; each refresh appended a new timestamped entry to the visible log, confirming writes were landing on the mounted volume.

```
(base) raghavdhoot@Raghavs-MacBook-Air-2 my-flask-app % docker ps
CONTAINER ID   IMAGE             COMMAND           CREATED         STATUS         PORTS
                NAMES
0b502afbc070   my-flask-app:v2   "python app.py"   2 minutes ago   Up 2 minutes   0.0.0.0:500
1->5000/tcp, [::]:5001->5000/tcp   flamboyant_gould
```

Stop and remove container — Stopped and removed the container entirely (docker stop + docker rm), leaving only the named volume behind.

```
Run 'docker run --help' for more information
(base) raghavdhoot@Raghavs-MacBook-Air-2 my-flask-app % docker run -d -p 5001:5000 -v my-app-
data:/app/data my-flask-app:v2
f4e82e56d2df04f43373c240b3267b5aecb221fe7b9d319b72d0bd8fa36d05ad
```

New container, same volume — Ran a fresh container from the same image, mounting the same my-app-data volume; Docker assigned a new container ID, confirming this is a completely separate container instance.

Hello, Docker!
Log:

```
2026-10-12 11:42:50: Hello, Docker!
2026-10-12 11:46:20: Hello, Docker!
2026-10-12 11:46:22: Hello, Docker!
2026-10-12 11:46:22: Hello, Docker!
2026-10-12 11:46:22: Hello, Docker!
2026-10-12 11:46:23: Hello, Docker!
```

Persistence verified — Accessed the app again — the log showed all prior entries from the deleted container plus new ones from this session, proving data persisted independently of container lifecycle via the named volume.